6-month Internship: Predicting the mean RF-EMF exposure around base stations from mobile traffic load and urban activity

Chair C2M, RFM², LTCI, Télécom Paris

Télécom Paris is a school of the IMT (Institut Mines-Télécom) and founding member of the Institut Polytechnique de Paris. Télécom Paris is recognized for its proximity to companies. Chair C2M, created in 2015, is a research chair in Télécom Paris. It focuses on the modelling, characterization and control of exposure to electromagnetic waves. It contributes through its research work to the characterization, analysis and modeling of exposure to electromagnetic waves induced by telecommunication systems.

Background: A base station radiates the power it needs to serve its users, so the radio frequency (RF) electromagnetic field (EMF) around it follows the traffic load it carries. At a commercial 5G mmWave site, the measured exposure grew in proportion to the downlink traffic [1], and exposure follows daily and weekly cycles [2].

Using this link to predict exposure is difficult because operators do not publish the load of their cells. Current predictors rely on static descriptors of the environment (buildings, road network, base station registry, population and building density [3]) and on learning models trained on probe and simulation data [4]. In the team's tests on a network of fixed probes in France [5], static descriptors explain the daily shape of the exposure but add little to the prediction of its mean level at a given site [6]. Over several years, exposure and traffic do not move together either: repeated outdoor measurements in four European countries found no significant change in base station exposure between 2016 and 2023, while mobile data traffic in Western Europe grew by a factor of eight [7].

Open data now allow the construction of proxies for the load seen by an antenna. The NetMob23 dataset gives the traffic of 68 mobile services on 100 m tiles every 15 minutes, for 77 days in spring 2019 over 20 French metropolitan areas. Earlier work links the daily signature of mobile traffic to land use [8]. Open census grids and mobility surveys describe where people live and where they are present during the day.

Which of these inputs improves the prediction of the mean exposure around a base station, beyond what static descriptors already give, and at which time scale, remains an open question. The evidence above points in different directions: the link is clear at the scale of hours and absent over years.

Objectives: The aim of this internship is to build a predictor of the mean exposure around a few base stations, and to find out which traffic-related and urban-activity inputs carry real predictive information. The work rests on the exposure time series of the team's fixed probes [5], on the base station registry and simulation maps of the ANFR, and on open traffic, population and land-use data.

In this internship, you will:

- Examine the literature summarised above with a critical eye: for each candidate input (traffic load proxies, population presence, base station registry and simulation maps, built-environment descriptors), state what the evidence supports and how to test it.
- Assemble the data and define the quantity to predict, the mean exposure around a base station (averaging window, probes attached to each station, missing values).
- Build predictors of increasing complexity, from simple baselines (city-level median, static descriptors, physical estimate) to models that add the load and activity proxies, and compare them with the neural-network model of the team [4].
- Evaluate on whole sites and periods left out of the training, with the uncertainty of the errors, and report which inputs improve the prediction and which do not, negative results included.

Requirements:

- Master's level in telecommunications, data science, applied mathematics or a related field
- Good in probability and statistics, experience in Python (pandas, scikit-learn or equivalent); knowledge of geospatial data and of mobile networks is a plus
- Good English and team working.

Practical Information: The duration of internship is 6 months between Feb. and Oct. 2027. Location is at Télécom Paris, 19 place Marguerite Perey, 91120 Palaiseau. For the application, please send your **CV** and **Transcripts** from last 3 years to:

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References

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